

VPS Setup Guide — Node.js Backend + Next.js Frontend (.next Build Copy Method)

1. Initial Server Setup

```
ssh root@SERVER_IP
```

```
apt update && apt upgrade -y
```

```
apt install -y curl git build-essential ufw
```

2. Install Node.js

```
curl -fsSL https://deb.nodesource.com/setup_20.x | bash -
```

```
apt install -y nodejs
```

```
node -v
```

```
npm -v
```

3. Install PM2

```
npm install -g pm2
```

```
pm2 startup
```

```
pm2 save
```

4. Install Nginx

```
apt install -y nginx
```

```
systemctl enable nginx
```

```
systemctl start nginx
```

5. Firewall

```
ufw allow OpenSSH
```

```
ufw allow 'Nginx Full'
```

```
ufw enable
```

6. Project Structure

```
/root/app/backend
```

```
/root/app/frontend
```

7. Backend Setup

```
cd /root/app/backend
```

```
npm install --production
```

```
pm2 start src/server.js --name backend
```

pm2 save

8. Next.js Setup Using Prebuilt .next Folder (No npm build on VPS)

On your LOCAL machine:

npm run build

tar -czf next-build.tar.gz .next

Copy build to server:

scp next-build.tar.gz root@SERVER_IP:/root/app/frontend/

On the SERVER:

cd /root/app/frontend

tar -xzf next-build.tar.gz --no-same-owner

Install only runtime deps:

npm install --production --no-audit --no-fund

Start Next.js server:

pm2 start npm --name frontend -- start

pm2 save

Important: package.json must contain:

```
"start": "next start -p 3001"
```

9. Nginx Config

Create file: /etc/nginx/sites-available/app.conf

```
server { listen 80; server_name example.com www.example.com; return 301 https://$host$request_uri; }
server { listen 443 ssl; server_name example.com www.example.com; ssl_certificate /etc/letsencrypt/live/example.com/fullchain.pem;
ssl_certificate_key /etc/letsencrypt/live/example.com/privkey.pem; location /api { proxy_pass http://localhost:5000;
proxy_set_header Host $host; proxy_set_header X-Real-IP $remote_addr; } location / { proxy_pass http://localhost:3001;
proxy_set_header Host $host; proxy_http_version 1.1; proxy_set_header Upgrade $http_upgrade; proxy_set_header Connection 'upgrade'; }
location /uploads/ { alias /var/www/uploads/; } }
```

Enable config:

ln -s /etc/nginx/sites-available/app.conf /etc/nginx/sites-enabled/

nginx -t

systemctl reload nginx

10. SSL Certificate

snap install core; snap refresh core

snap install --classic certbot

certbot --nginx -d example.com -d www.example.com

11. Updating Frontend Later

Build locally → tar .next → scp → extract → pm2 restart frontend

12. Safe Extract Command (Low Memory VPS)

tar -xzf next-build.tar.gz --no-same-owner